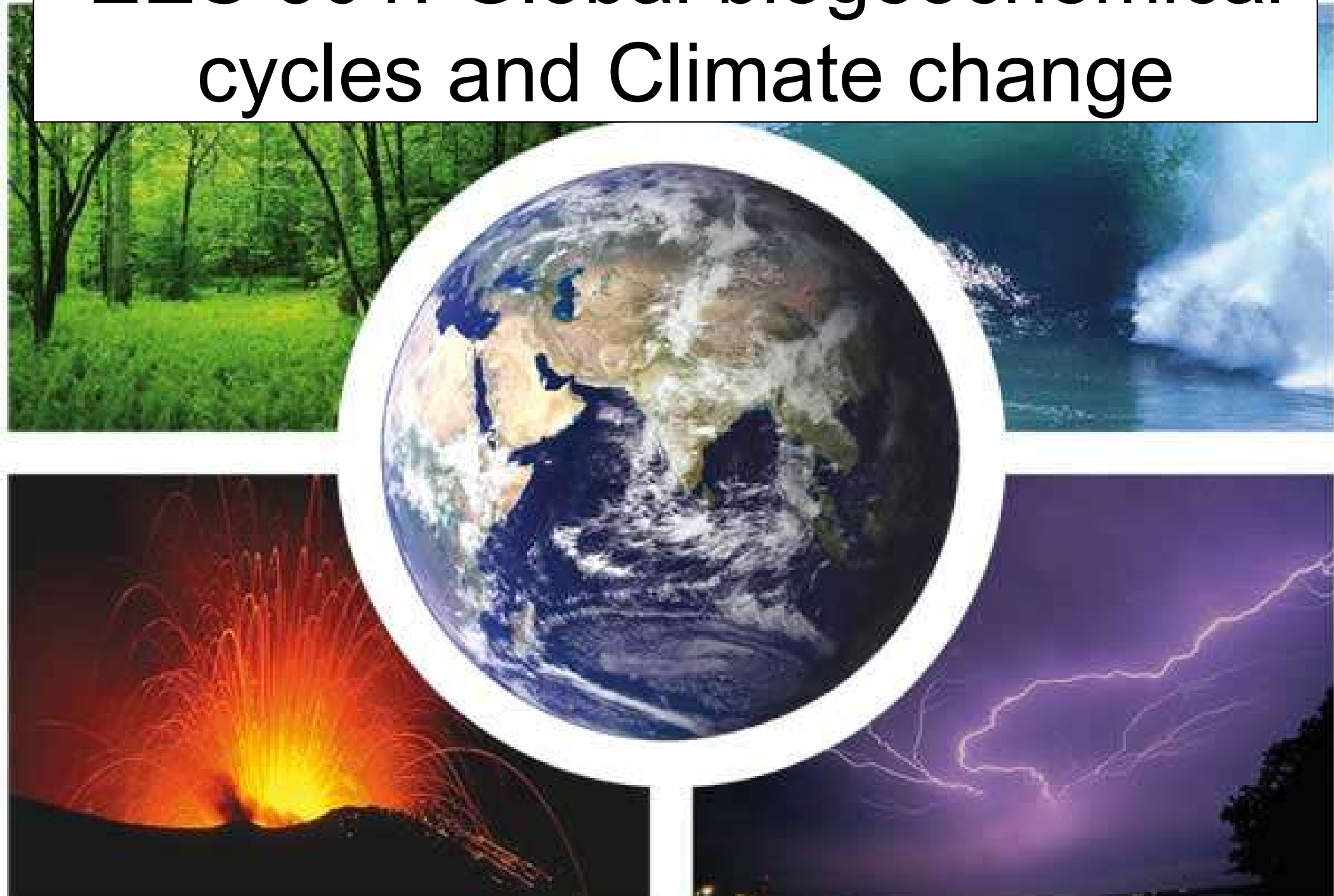
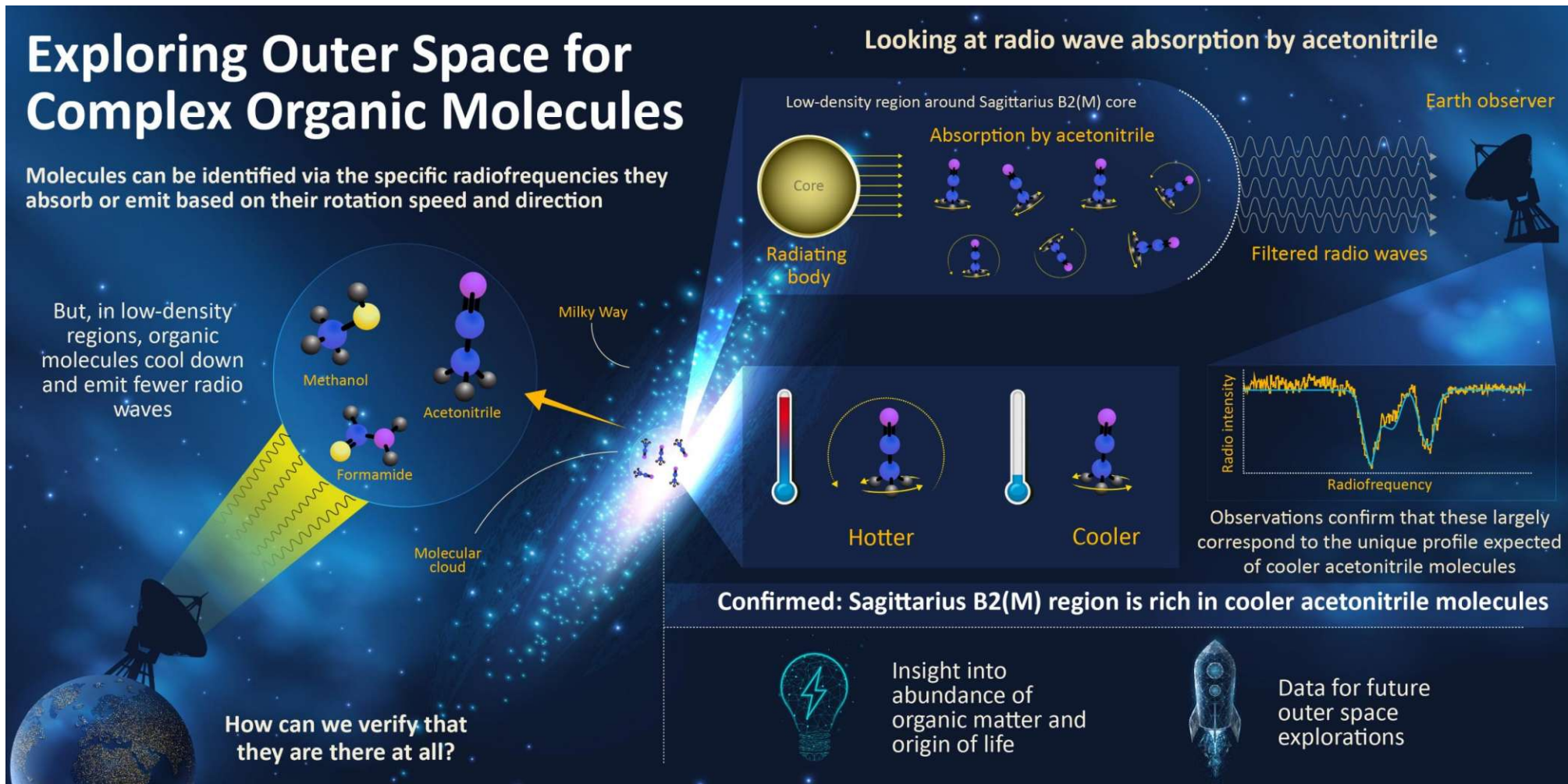


EES 301: Global biogeochemical cycles and Climate change



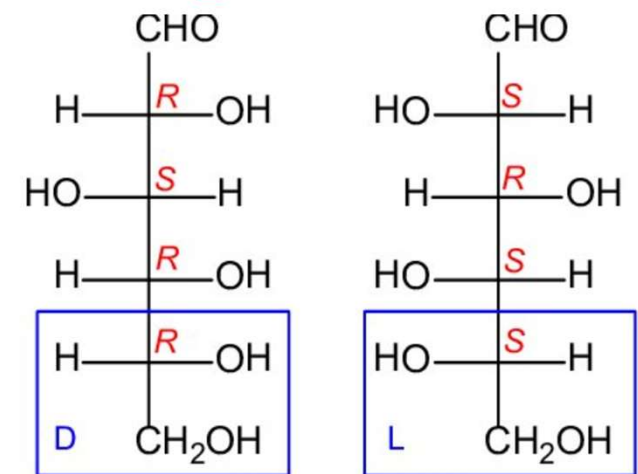
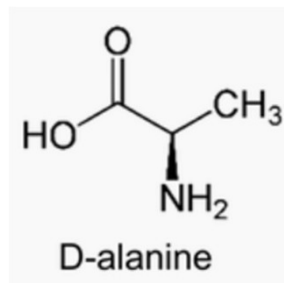
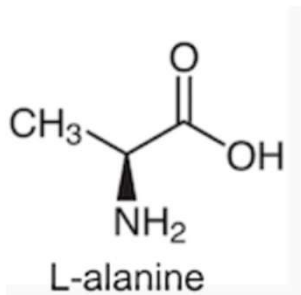
Astrobiology & The matter planets are made up of:

Carbon containing molecules, from carbon monoxide to polycyclic aromatic hydrocarbons, are detected in star systems



The origin of life on Earth

1. Life is primarily made up of hydrogen carbon, oxygen, nitrogen, phosphorous and sulfur
2. Life is believed to have developed from the **abiotic building blocks of life, complex organic molecules**
3. Our present views on the origin of life are based on **meteorites, imaginative experiments in the lab and deductions from surviving genetic code**
4. **It is possible that meteorites delivered the basic building blocks of life namely sugars, organic acids and amino acids to earth**
5. **The abiotic organic molecules found in meteorites are mostly racemic. How life settled on L amino acids and D sugars remains a mystery**



Material available in interstellar space

Molecules in Sagittarius cloud



➤ The existence of “simple” organic compounds in the interstellar cloud has been proven by radiowave spectroscopy

Sagittarius cloud



➤ Carbonaceous meteorites contain organic matter formed intergalactic space. The organic matter includes sugars, organic acids and amino acids

➤ The fact that the amino acids in the meteorites mostly contain equal proportion of D and L stereo-isomers indicates they were produced abiotically and are not contaminated by terrestrial amino acids

➤ More than 100 different amino acids have been detected in meteorites. Strecker-type syntheses for α -amino acids or Michael addition for β -amino acids, or the secondary processes on the parent body appear to have generated them

➤ prebiotic sugar formation reaction via the “formose reaction” may have used gamma radiation of differentiation heat as energy source with the meteorite serving as catalyst

Carbonaceous chondrite



Early attempts at abiotic synthesis

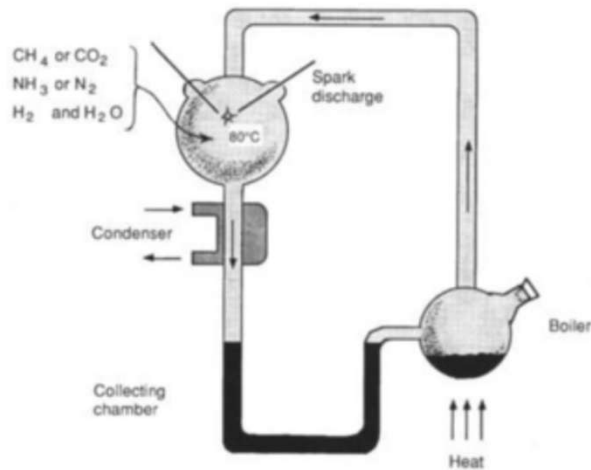
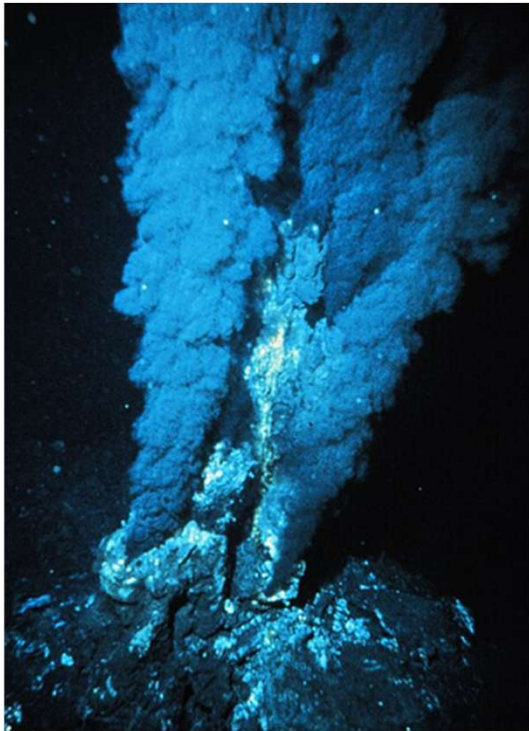


FIG. 3-1 Schematic of the apparatus Stanley Miller used to demonstrate formation of amino acids from simple inorganic compounds under conditions similar to those of the early Earth.



➤ Stanley Miller in 1953 produced amino acids abiotically in a closed flask containing water and an atmosphere comprising of methane, ammonia and hydrogen with lightening flashes as energy source.

➤ While the experiments were successful, this type of synthesis is more likely on objects beyond the Kuyper belt than on Earth.

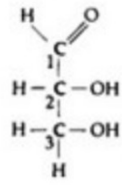
➤ The atmospheric composition was inspired by the types of atmospheres found in the outer planets but we are very certain that terrestrial planets had a CO_2 rich atmosphere.

➤ The energy source used by Miller is also very destructive for life as we know it. Today we believe that hydrothermal vents on the ocean floor were the more likely energy source for the evolution of life.

➤ It is noteworthy that hydrothermal alteration of the parent body is associated with some amount of L- amino acid excess in meteorites. The amino acids in those meteorites were not terrestrial contamination. Their Deuterium and ^{13}C excess proves that they originated in the outer parts of the solar system.

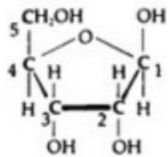
different monosaccharides

3-CARBON SUGAR

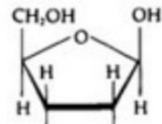


Glyceraldehyde

5-CARBON SUGARS

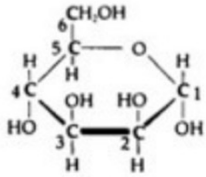


Ribose

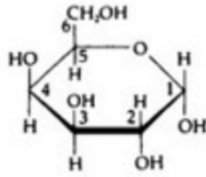


Deoxyribose

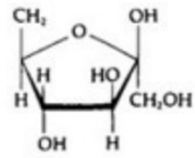
6-CARBON SUGARS



α-Mannose

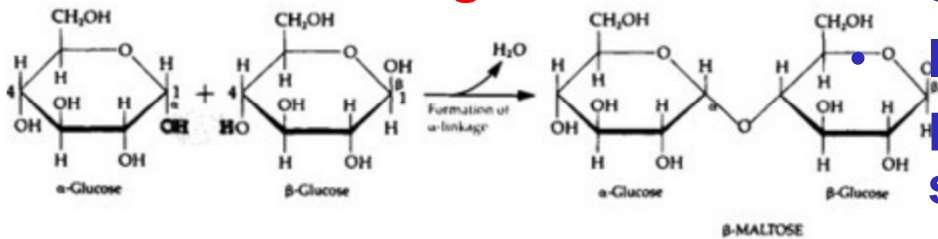


α-Galactose

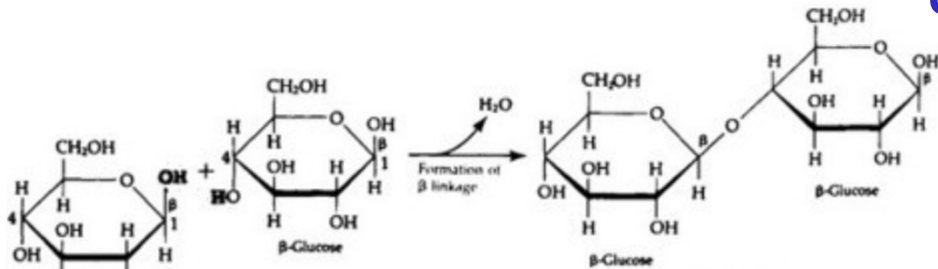


Fructose

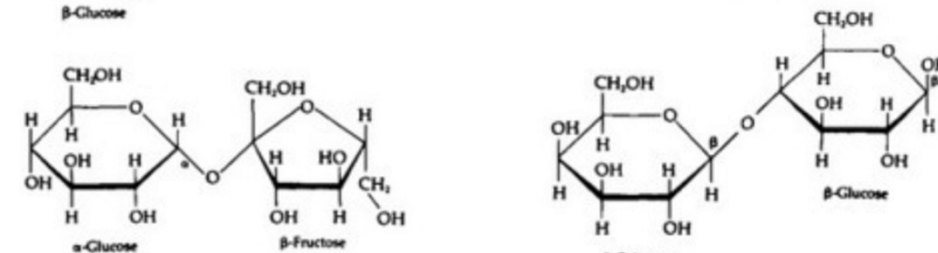
(a) different oligosaccharides



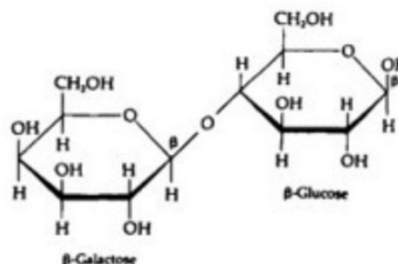
β-MALTOSE



CELLOBIOSE



SUCROSE

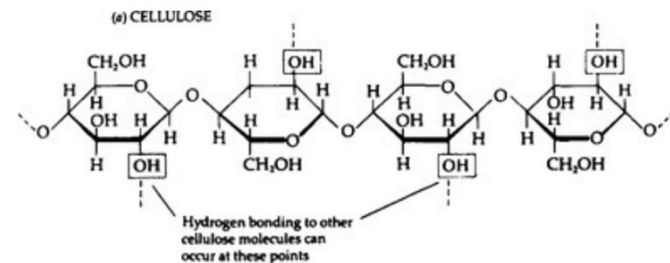


β-LACTOSE

The chemical basis of life

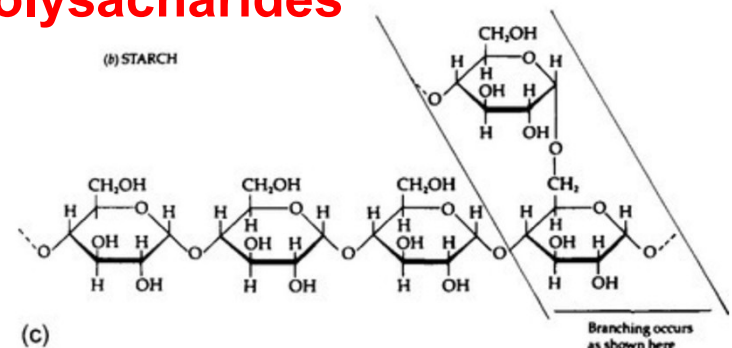
- Carbohydrates form a diverse group of compounds that share an approximate formula $(CH_2O)_n$
- They are classified into monosaccharides, oligosaccharides and polysaccharides
- Derivative carbohydrates such as sugar phosphase and amino sugars contain additional elements

Important amino sugars are chitin the principal structural carbohydrate in insect skeletons and the cell walls of fungi and cartilage



Hydrogen bonding to other cellulose molecules can occur at these points

different polysaccharides

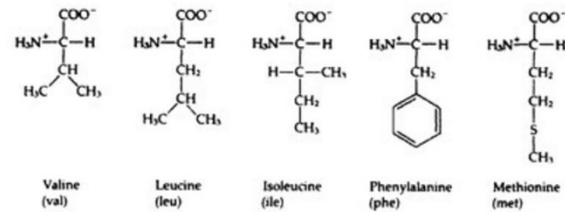


Branching occurs as shown here

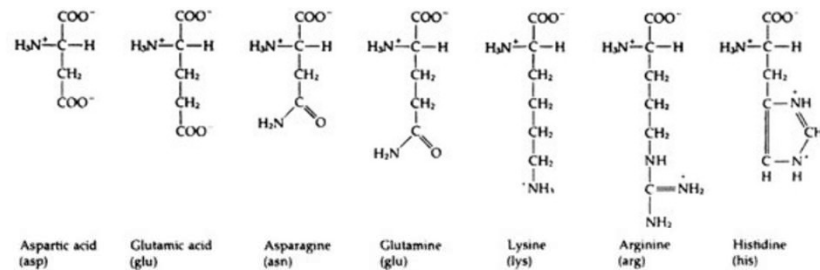
The chemical basis of life – amino acids and proteins

20 AMINO ACIDS FOUND COMMONLY IN PROTEINS

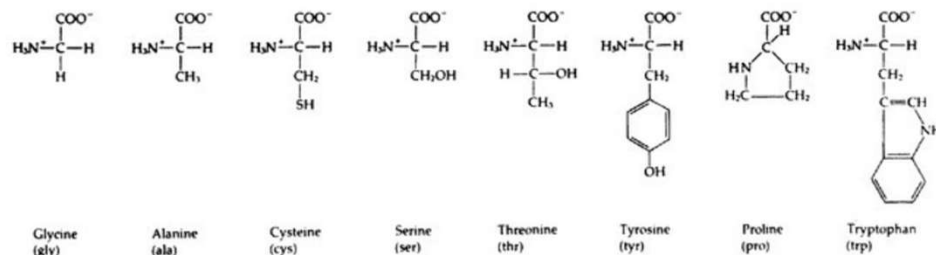
A. AMINO ACIDS WITH HYDROPHOBIC R GROUPS



B. AMINO ACIDS WITH HYDROPHILIC R GROUPS



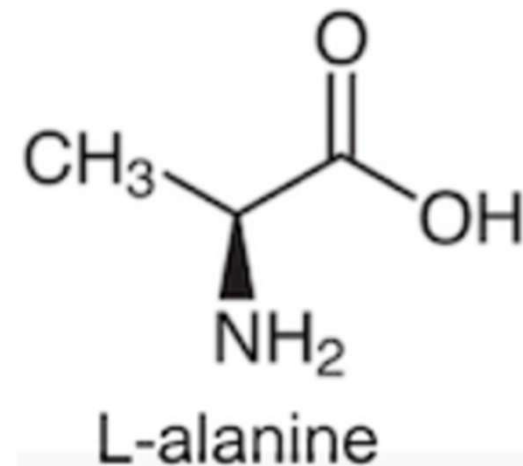
C. AMINO ACIDS THAT OCCUR BOTH ON THE SURFACE AND IN THE INTERIOR OF PROTEINS



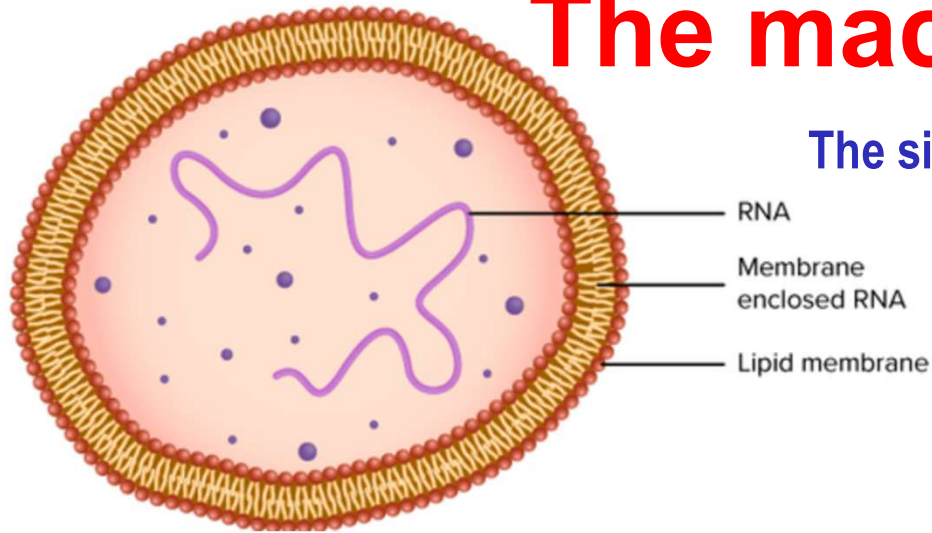
- Amino acids the building blocks of giant protein molecules have a carboxyl group and an amino group attached to the same carbon
- A protein is a linear polymer of amino acids combined by peptide linkages
- Twenty different amino acids are common in proteins. Why those 20 among the 100s discovered in meteorites form the basis of life needs to be explained under the “origin of life” theory

FIG. 3-10 The 20 protein amino acids divided by R group character as (a) hydrophobic, (b) hydrophilic, and (c) mixed.

(Reprinted with permission from W. K. Purves and G. H. Orians, “Life: The Science of Biology,” pp. 63–81, Copyright © 1987 by Sinauer Associates, Inc., Sunderland, MA.)



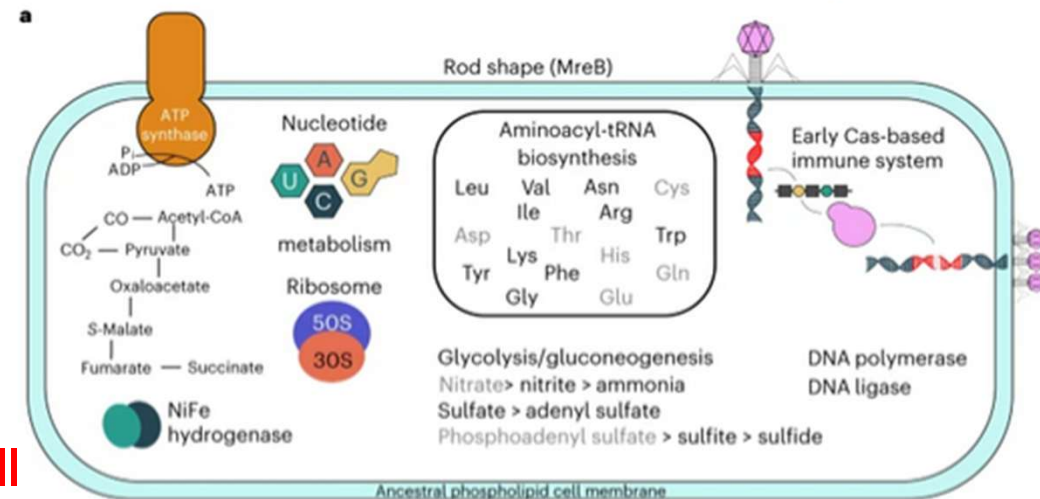
The machinery of life



The simplest requirements for life as we know it are:

- The presence of a semi permeable membrane to control the passage of materials into and out of compartments that we call “cells”
- Chemical machinery for the synthesis and degradation of essential molecules
- Genetic material that encodes the synthesis of the molecules required to catalyze those syntheses and degradations
- Sufficient structure to prevent unwanted reactions from occurring
- A machinery to duplicate all those capabilities

Fig. 3: A reconstruction of LUCA, within its evolutionary and ecological context.



The last universal common ancestor (LUCA) is generally studied with genetic tools. A 2024 article in Nature Ecology & Evolution claimed LUCA lived ~4.2 Ga (4.09–4.33 Ga) and suggests that LUCA had a genome of at least 2.5 Mb (2.49–2.99 Mb), encoding around 2,600 proteins, comparable to modern prokaryotes.

<https://www.nature.com/articles/s41559-024-02461-1>

LUCA is not the first bacterial cell
LUCA is just the ancestor of everything that survived

The chemical basis of life - DNA

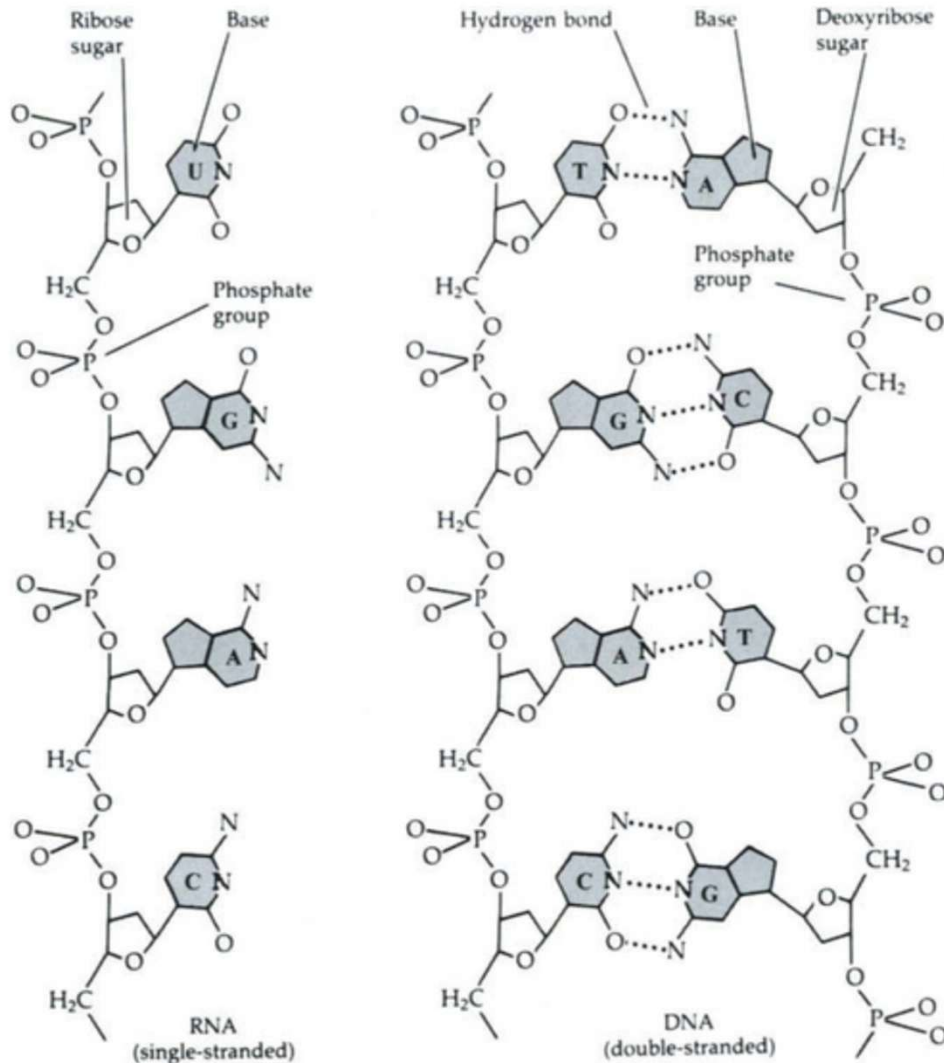


FIG. 3-13 RNA and DNA, the carriers of genetic information. Bases are denoted as A = adenine, T = thymine, G = guanine, C = cytosine, and U = uracil. Note that RNA contains U where DNA contains T. (Reprinted with permission from W. K.

Purves and G. H. Orians, "Life: The Science of Biology," pp. 63-81, Copyright © 1987 by Sinauer Associates, Inc., Sunderland, MA.)

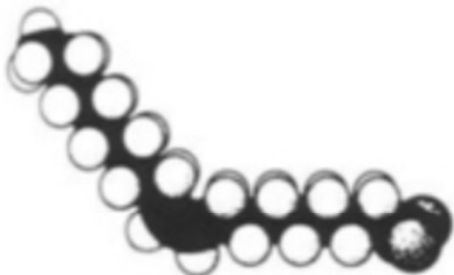
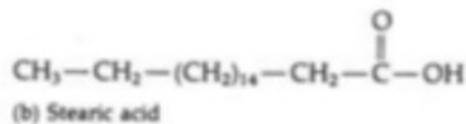
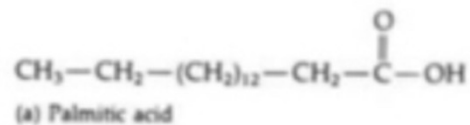
- Deoxyribonucleic acid DNA is the genetic material of all organisms including plants, animals and microorganisms.
- Some viruses lack DNA but use RNA (ribonucleic acid) in its place.
- DNA carries all the hereditary information of the organism. It is replicated and passed from parent to offspring.
- Today RNA is formed on a DNA template in the nucleus of a cell. However RNA may predate DNA
- The RNA carries the genetic information to the cytoplasm where it is used to produce proteins on the ribosome.
- The specific proteins formed include enzymes which carry out the characteristic activities of the organism.
- Both RNA and DNA are formed from monomers called nucleotides each of which consists of a simple sugar, a phosphate group and a nitrogen containing base.
- The genetic alphabet contains only four letters.

The chemical basis of life - lipids

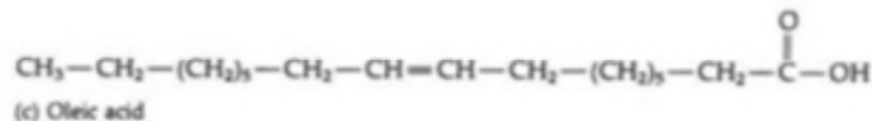
- Lipids are insoluble in water and release large amounts of energy when they are metabolized. **Lipids are not produced abiotically.**
- The lipids of Eubacteria and Eucarya are composed of two principal building blocks, **fatty acids and glycerol**. Three fatty acids (carboxylic acids with long hydrocarbon tails) combine with one molecule of glycerol by ester linkages to form a triglyceride
- More complex lipids are formed by the addition of other groups the most important of which contains phosphorus.
- Archaea synthesize membranes of isoprenoid lipids**



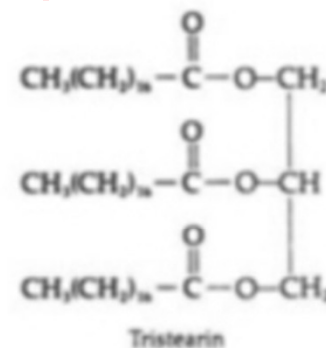
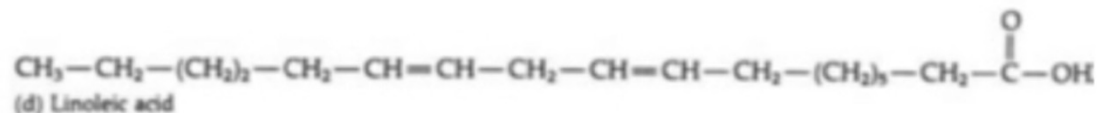
Model of palmitic acid



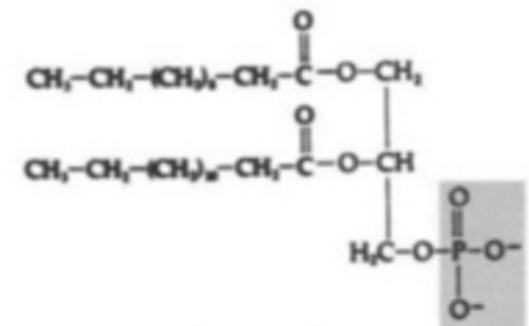
Model of oleic acid



(a)



TRIGLYCERIDE (A FAT)



(u) Phosphatidate

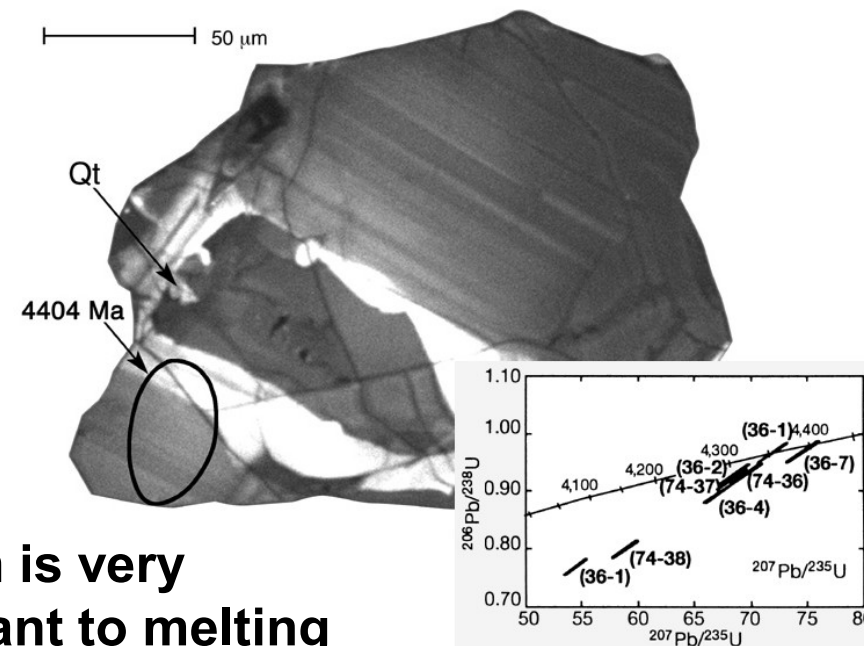
Geological evidence for early life

- The earliest evidence for life on Earth is graphite found to be biogenic in 3.7-billion-year-old metasedimentary rocks discovered in Western Greenland <https://www.nature.com/articles/ngeo2025>
- **Microbial mat fossils found in 3.48-billion-year-old sandstone discovered in Western Australia.**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3870916/pdf/ast.2013.1030.pdf>
- **In 2015, possible remains of biotic matter were found in 4.1-billion-year-old rocks in Western Australia. “Potentially biogenic carbon preserved in a 4.1 billion-year-old zircon”**
<https://www.pnas.org/doi/pdf/10.1073/pnas.1517557112>

The Earth's oldest mineral grain in a sandstone in Jack Hill is 4.404 Ga old. Since plate tectonics recycles materials the oldest records have not survived



Fossil microbial mats



Zircon is very resistant to melting

The chemical basis of life – ATP in the energy metabolism

- To build their structures and carry out a myriad of biochemical reactions, cells need an energy source.
- The energy is obtained via biochemical pathways driven by sunlight or by energy contained in reduced chemical compounds.

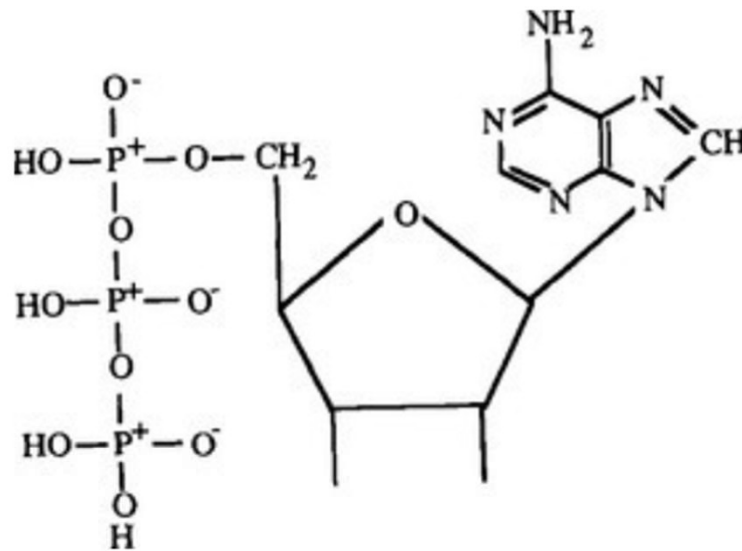


FIG. 3-2 Chemical diagram of ATP (adenosine triphosphate). The three functional groups are the base adenosine (upper right), a five-carbon ribose sugar (middle), and three molecules of phosphate (left). Lines at bottom of sugar ring indicate hydroxyl groups.

- Adenosine triphosphate (ATP) is the key energy currency in all life as we know it.
- Hence the ability to synthesize this compound and the genetic code for ATP synthase must be one of the oldest pieces of genetic code.
- Most organo-phosphorus compounds are phosphates $R-PO_4$. The bond energy of the phosphate to the rest of the molecule is central to the flow of energy in all metabolic processes.

Energy metabolism - glycolysis

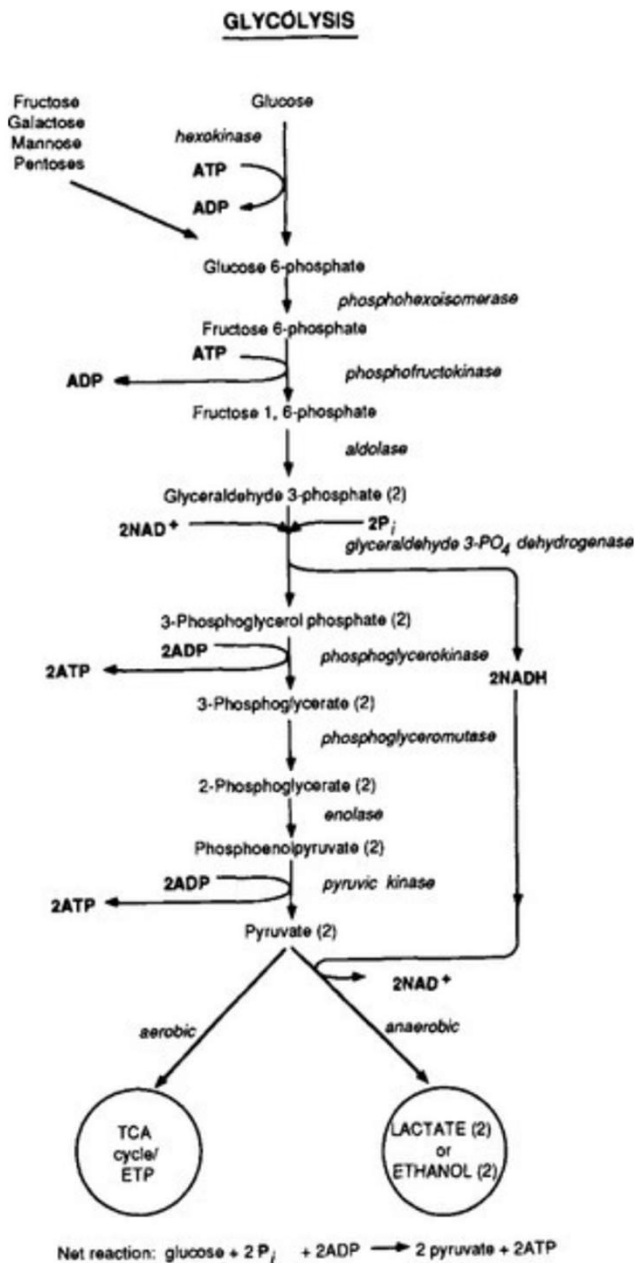


FIG. 3-3 The biochemical pathway of glycolysis, which obtains energy from the breakdown of 6-carbon sugars to a pair of 3-carbon pyruvate molecules. Enzymes at each reaction in the sequence are in *italics*. The energy generated is stored in the form of ATP (see [Fig. 3-2](#)).

- The ability of driving under anaerobic conditions by deriving energy from carbohydrates via some form of fermentation is also an ancient form of the energy metabolism
- In this process sugars are first oxidized to form pyruvic acid in this process ATP is produced during one of the steps.
- Glycolysis participates in both bioenergetics and building block syntheses in cells.
- It is a universal, but not uniform pathway in all known domains of contemporary living beings.
- The lower reaction chain, named trunk pathway, is highly conserved. A very small number of variations are only observed, indicating its ancient origin. This is believed to have been present in LUCA

Energy metabolism – chemoautotrophs

- Some anaerobic bacteria and archaea use substances such as nitrate, iron and manganese oxides, sulfates and carbon dioxide as an ultimate electron acceptor during respiration
- Such microbes are common in sediments, extreme environments and around smokers on the ocean floor. Many early microbes of this type are believed to have gone extinct during the oxygen holocaust also known as great oxidation event (GOE).
- Some metabolic pathways such as methanogenesis, and photoferrotrophy are ancient and believed to have dominated some biogeochemical cycles before the great oxidation event.
- Iron, manganese and sulfur oxidation and reduction reaction pathways also originated in the early Achaean.

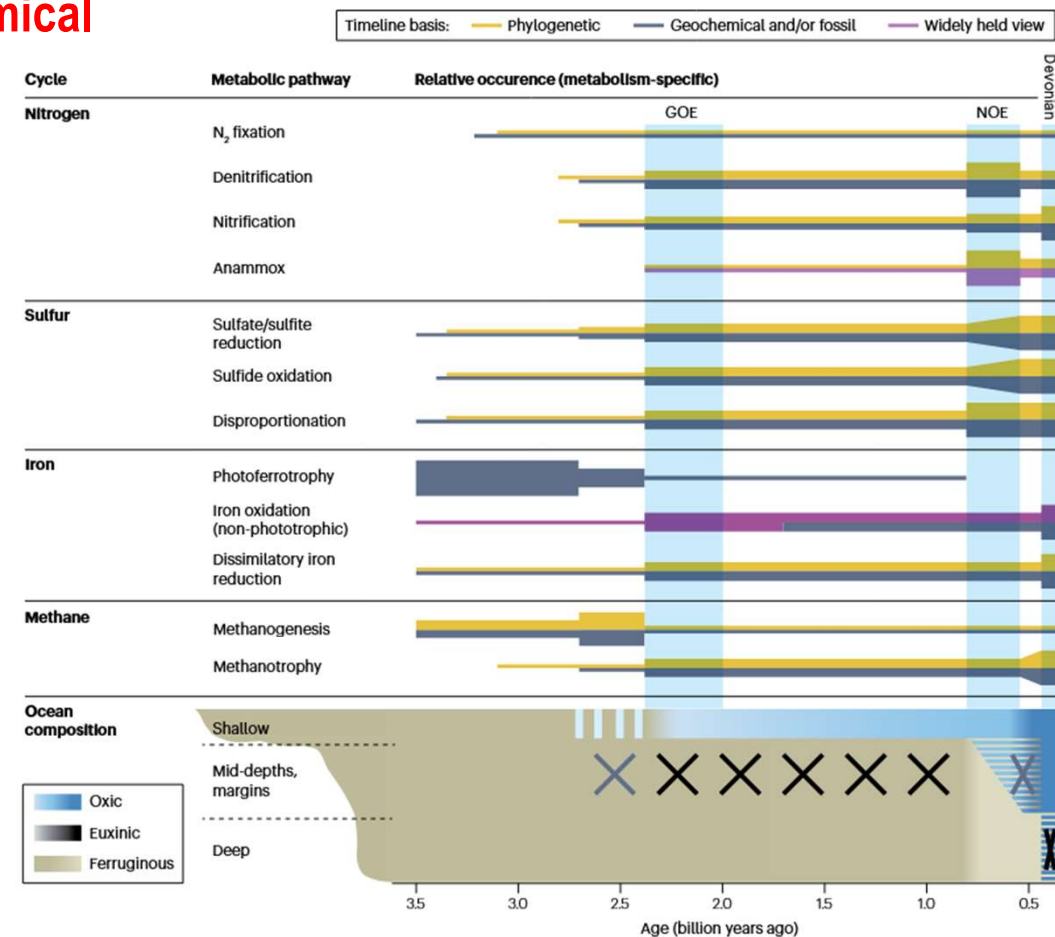
Table 3-1

Electron acceptors that are used in the biodegradation of organic material in marine sediments. More on the chemistry of these processes is presented in [chapter 8](#) and [16](#)

Electron acceptor	Process	Approximate depth in sediment (cm)
O ₂	Aerobic respiration	0–0.5 ^a
FeO(OH)	Iron respiration	0–0.5 ^a
NO ₃ ⁻	Denitrification	0.5–5 ^a
SO ₄ ²⁻	Sulfate reduction	5–100 ^b
CO ₂	Methanogenesis	> 100 ^b

^aShallow (possibly water column).

^bDeeper (sediment).



Evolution of diversity

- The main difference between archaea and bacteria are differences in the structure of the cell membrane and metabolism. Methanogens, Thermophiles, and Halophiles are archaea
- The evolution of the Eukaryotic cell 2.7 to 1.8 Ga ago presents a milestone in the evolution of life. Protecting the genetic code with an extra membrane protected it from chemical reactions.

